

Md Nazmul Islam

📍 Sirajganj, Bangladesh

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Research Interests

Composite Materials • Computational Mechanics • Finite Element Analysis • Additive Manufacturing
• Machine Learning • Structural Integrity and Failure Analysis

Education

Khulna University of Engineering & Technology (KUET)

Khulna, Bangladesh

Bachelor of Science in Mechanical Engineering

Dec 2016 – Apr 2022

CGPA: 3.45/4.00

Selected Coursework:

- Finite Element Method
- Fatigue, Creep, and Fracture
- Simulation and Modeling
- Mechanics of Composite Materials
- Heat Transfer
- Machine Design

Research Experience

Undergraduate Thesis Research

KUET

Computational Study of Multi-layer Printed Circuit Boards Under Bending Load

Advisor: Dr. Md. Shariful Islam

- Developed finite element models of multilayer PCB structures under three-point bending using ABAQUS.
- Investigated the effects of laminate thickness and stacking sequence on stress distribution and bending stiffness.
- Evaluated structural response and mechanical behavior of composite laminate systems under bending load conditions.

Independent Research and Skill Development

2022 – Present

- Conducting independent research focused on composite materials, computational mechanics, and manufacturing technologies.
- Expanding expertise in finite element analysis, numerical simulation, and simulation-driven engineering design.
- Studying machine learning applications in manufacturing, predictive modeling, and material behavior analysis.
- Strengthening research capabilities through scientific literature review, software training, and technical project development.

Publications

Islam, M.N.*, Anwar, M.S., Islam, M.S., Arifuzzaman, M., and Al Bari, M.A.

Bending Analysis of Glass Fiber Reinforced Epoxy Composites/Copper-Clad Laminates for Multi-Layer Printed Circuit Boards.

Hybrid Advances, Vol. 4, 100090, 2023.

DOI: [10.1016/j.hybadv.2023.100090](https://doi.org/10.1016/j.hybadv.2023.100090)

**First Author*

Projects

Design and Fabrication of Manual Die-Cutting Machine

Supervisor: Dr. Md. Kutub Uddin

- Designed and fabricated a low-cost, manually operated die-cutting machine for small-scale applications.
- Performed structural analysis and component selection to improve operational reliability and cutting precision.

Technical Skills

- | | |
|---|--|
| • Computational Mechanics
ABAQUS, Finite Element Analysis (FEA) | • CAD and Engineering Design
SolidWorks, AutoCAD |
| • Programming and Numerical Computing
Python, MATLAB, C | • Data Analysis and Visualization
OriginPro, NumPy, Matplotlib |

Professional Experience

Operations and Production Management

2022 – Present

Family Manufacturing Business (Twisting Mill)

- Supervise daily manufacturing operations, workforce management, and production activities.
- Diagnosed and resolved machinery-related problems involving shaft fatigue, vibration, and component failure.
- Applied mechanical engineering principles to practical industrial maintenance and troubleshooting challenges.

Training

- Training in AutoCAD and SolidWorks, CADers, KUET (2017)
- Programming and Introduction to Robotics Workshop, LOOP, KUET (2018)
- Training in CNC, CAM, and 3D Printing, KUET (2019)

Certifications

- IELTS Academic, British Council (Overall Band Score: 7.0)
- Introduction to Programming with MATLAB – Vanderbilt University (Coursera)
- Excel Skills for Business (Essential to Advanced) – Macquarie University (Coursera)

Awards and Honors

- University Technical Scholarship (Four Consecutive Years)

Industrial Activities

- Industrial Visit to Chittagong Dry Dock Ltd., Bangladesh
- Industrial Visit to Eastern Refinery Ltd., Bangladesh
- Industrial Visit to Port of Chittagong, Bangladesh

References

Dr. Md. Shariful Islam

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Dr. Md. Arifuzzaman

Professor, Department of Mechanical Engineering, KUET

Email: arif48@me.kuet.ac.bd

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